

CHE 201 Spring 2020 Midterm 2

Anna Christensen

TOTAL POINTS

52 / 100

QUESTION 1

Question 1 25 pts

1.1 A 9 / 21

- 0 pts Correct
- ✓ - 3 pts first one is incorrect
- 3 pts second one is incorrect
- 3 pts third one is incorrect
- ✓ - 3 pts fourth one is incorrect
- 3 pts fifth one is incorrect
- ✓ - 3 pts sixth one is incorrect
- ✓ - 3 pts seventh one is incorrect

1.2 B 0 / 4

- 0 pts Correct
- ✓ - 4 pts not done or incorrect

QUESTION 2

Question 2 40 pts

2.1 a 1 / 10

- 0 pts Correct
- ✓ - 3 pts Qc not labeled or wrong direction
- ✓ - 3 pts Qh not labeled or wrong direction
- ✓ - 3 pts work not labeled or wrong direction

2.2 b 0 / 5

- 0 pts Correct
- ✓ - 5 pts incorrect or not done
- 2 pts only one calculated
- 2 pts partially done/not calculated for year
- 2 pts without cooling is incorrect

2.3 C 0 / 5

- 0 pts Correct
- ✓ - 5 pts not done/incorrect

- 1 pts calculation error

2.4 d 2 / 5

- 0 pts Correct
- 5 pts incorrect/not done
- ✓ - 3 pts Th is taken as 90C

2.5 e 8 / 10

- 0 pts Correct
- ✓ - 2 pts calculation error
- 10 pts not done/incorrect
- 5 pts only one calculated
- 2 pts unit conversion

2.6 f 4 / 5

- 0 pts Correct
- 5 pts incorrect /reverse formula/ not done
- 1 pts reversible/irreversible?
- ✓ - 1 pts calculation error

QUESTION 3

Question 3 35 pts

3.1 a 5 / 5

- ✓ - 0 pts Correct
- 3 pts two outlets are not shown
- 2 pts inlet is not shown
- 1 pts boundary is not labeled

3.2 b 5 / 5

- ✓ - 0 pts Correct
- 2 pts open system not stated
- 2 pts steady state is not stated
- 1 pts delta PE is zero

3.3 C 3 / 5

- 0 pts Correct
- 5 pts incorrect
- ✓ - 2 pts work can not be canceled

3.4 d 8 / 10

- 0 pts Correct
- 2 pts h2 is wrong
- 2 pts KE term is not divided by 1000 (unit conversion)
- 10 pts incorrect or not done
- 2 pts not multiplied with mass outlets
- 1 pts work sign is wrong
- ✓ - 2 pts work is not added
- 2 pts work unit is not converted
- 2 pts mass is wrong
- 2 pts h1 is wrong
- 2 pts h3 is wrong
- 0 pts Click here to replace this description.

3.5 e 3 / 5

- 0 pts Correct
- 5 pts incorrect/not done
- ✓ - 2 pts specific volume is wrong

3.6 f 4 / 5

- 0 pts Correct
- 2 pts v2 is wrong
- 2 pts state 1 not labeled
- 1 pts v3 is wrong
- 5 pts not done /wrong
- 2 pts dome missing
- ✓ - 1 pts state 3 is wrong
- 1 pts v1 is wrong

①
a) False

True

False

True

False

False

True

$$T_c = 100 + 73 = 373 \quad T_H = 200 + 273 = 473$$

$$\frac{T_c}{T_H - T_c} = \frac{373}{473 - 373} = 3.73$$

b)

$$R_c = \frac{QD}{VA} = \frac{AVD}{VA} = \frac{VD}{V}$$

$$\frac{R_c}{VD} = \gamma$$

1.1 A 9 / 21

- 0 pts Correct

✓ - 3 pts first one is incorrect

- 3 pts second one is incorrect

- 3 pts third one is incorrect

✓ - 3 pts fourth one is incorrect

- 3 pts fifth one is incorrect

✓ - 3 pts sixth one is incorrect

✓ - 3 pts seventh one is incorrect

① False

a)

True

False

True

False

False

True

$$T_c = 100 + 73 = 373 \quad T_H = 200 + 273 = 473$$

$$\frac{T_c}{T_H - T_c} = \frac{373}{473 - 373} = 3.73$$

b)

$$R_e = \frac{QD}{VA} = \frac{AVD}{VA} = \frac{VD}{V}$$

$$\frac{R_e}{VD} = \gamma$$

1.2 B 0 / 4

- 0 pts Correct

✓ - 4 pts not done or incorrect

2) rect. cycle

$$T_{avg} = 25^\circ C = 298 K$$

$$T_H = 90^\circ C = 363 K$$

$$T_C = 4^\circ C = 277 K$$

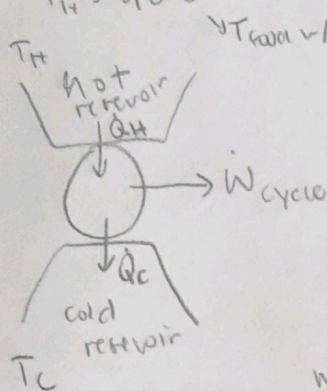
$$S = 3.5 \frac{kJ}{kg^\circ C}$$

4 kg

3x a week

$$\beta = 1.7$$

a)



(b)

$$W = Q_H - Q_C$$

$$\beta = \frac{Q_C}{W_{cycle}} = \frac{Q_C}{Q_H - Q_C} = 1.7$$

reversible: Q_{ref}

$$T_{H, without cook} = 25^\circ C$$

$$T_{H, with cook} = 90^\circ C$$

$$T_C = 4^\circ C$$

$$Q = T(S)$$

w/o cook

$$Q_{H1} = 25^\circ C (3.5 \frac{kJ}{kg^\circ C}) = 87.5 \frac{kJ}{kg} \times 4 kg = 350 kJ$$

with cook

$$Q_{H2} = 90^\circ C (3.5 \frac{kJ}{kg^\circ C}) = 315 \frac{kJ}{kg} \times 4 kg = 1260 kJ$$

with out cook

with cook

$$1.7 = \frac{Q_C}{315 - Q_C}$$

$$315 - Q_C$$

$$1.7 = \frac{Q_C}{87.5 - Q_C} \rightarrow 148.75 - 1.7 Q_C = Q_C$$

refrigeration cycle receives Q_H

$$Q_C = 55.09 \frac{kJ}{kg}$$

without cooking

$$W = 32.41 \frac{kJ}{kg}$$

c)

$$\text{with cooking } Q_C = 198.3 \frac{kJ}{kg}$$

$$W = 315 - 198.3 = 116.7 \frac{kJ}{kg}$$

$$\text{without cooking } Q_C = 55.09 \frac{kJ}{kg}$$

$$W = 87.5 - 55.09 = 32.41 \frac{kJ}{kg}$$

without cooking

$$\text{d) } \beta_{max} = \frac{T_C}{T_H - T_C} = \frac{277 K}{363 K - 277 K} = 3.22$$

$$\frac{277}{298 - 277} = 13.2$$

$$116.7 \frac{kJ}{kg} \times 4 kg = 466.8 kJ \text{ with } W_{cook}$$

$$32.41 \frac{kJ}{kg} \times 4 kg = 129.64 kJ \text{ with } W_{out cook}$$

total energy rejected = $Q_C + W$

$$\text{e) } 24 \text{ hr} \times 7 = 168 \text{ hr in week} \times \frac{1}{3 \text{ times a week}} = 56 \text{ hr}$$

$$\text{w/ cook } \frac{466.8 kJ}{56 \text{ hr}} \times \frac{1 \text{ hr}}{3600 \text{ s}} = 0.0023 \text{ kW}$$

$$0.0023 \text{ kW} \times 8760 \text{ hr} = 20 \text{ kWh}$$

$$365 \times 24 = 8760 \frac{\text{hr}}{\text{year}}$$

$$20 \text{ kWh} \times \frac{12 \text{ cents}}{\text{kWh}} = 242 \text{ cents} = \$2.41$$

$$\textcircled{C} \quad \frac{129.64 \text{ kJ}}{56 \text{ hr}} \times \frac{1 \text{ hr}}{3600 \text{ s}} \times 8760 \text{ hr} \times \frac{12 \text{ cents}}{\text{kWh}} = 67.59 \text{ cents}$$

$$= \$0.68$$

$$2.41 - 0.68 = \boxed{\$1.73 \text{ saved}}$$

$$\textcircled{F} \quad \oint \left(\frac{\delta Q}{T} \right) = -\epsilon_{\text{cycle}} = \frac{Q_{\text{in}}}{T_H} - \frac{Q_{\text{out}}}{T_C}$$

$$Q_{\text{in}} = Q_H = 87.5 \frac{\text{kJ}}{\text{kg}} \rightarrow 298 \text{ K}$$

$$Q_{\text{out}} = Q_C = 315 \frac{\text{kJ}}{\text{kg}} \rightarrow 363 \text{ K}$$

$$Q_{\text{out}} = Q_C \rightarrow 55.09$$

$$198.3 \rightarrow T_C = 277 \text{ K}$$

$$-\epsilon_{\text{cycle}} = \frac{87.5}{298} - \frac{55.09}{277} = 0.09 \rightarrow \epsilon = -0.09 \quad \text{impossible}$$

$$-0.09 < 0$$

$$-\epsilon_{\text{cycle}} = \frac{315}{363} - \frac{198.3}{277} = -6.46 \rightarrow \epsilon = 6.46 \quad \text{possible}$$

$$6.46 > 0$$

there are irreversibilities present within the system

2.1 a 1 / 10

- 0 pts Correct

- ✓ - 3 pts Qc not labeled or wrong direction
- ✓ - 3 pts Qh not labeled or wrong direction
- ✓ - 3 pts work not labeled or wrong direction

2.2 b 0 / 5

- 0 pts Correct

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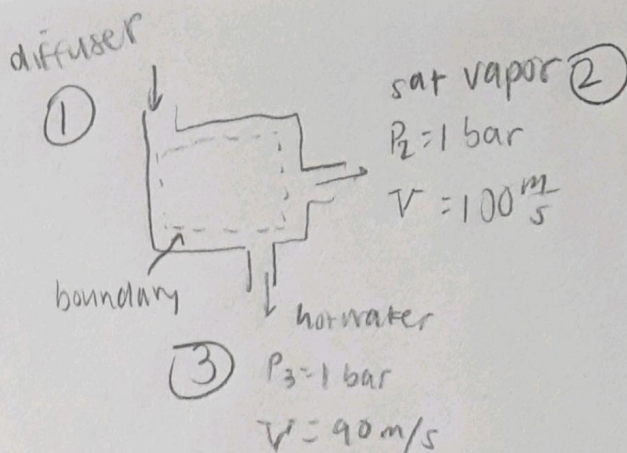
2.6 f 4 / 5

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③ a) $P_1 = 30 \text{ bar}$
 $T_1 = 400^\circ\text{C}$
 $V_1 = 40 \frac{\text{m}}{\text{s}}$
 $\dot{m} = 2 \frac{\text{kg}}{\text{s}}$

Power out = 2 MW

2 outlets



b) Engineering model

- open system
- inlet, 2 outlet
- steady state
- $PE = 0$
- $\dot{W}_{cv} = 0$

d) $\dot{Q}_{cv} = ?$

$$-\dot{Q}_{cv} = 2 \frac{\text{kg}}{\text{s}} \left[\right]$$

$$h_1 = 3230.9 \frac{\text{kJ}}{\text{kg}}$$

$$P_1 = 30 \text{ bar}$$

$$T_1 = 400^\circ\text{C}$$

$$V_1 = 0.0994 \frac{\text{m}^3}{\text{kg}}$$

$$h_2 = \text{sat vapor}$$

$$2675.5 \frac{\text{kJ}}{\text{kg}}$$

$$h_3 = 417.46$$

$$V_2 = 1.694$$

$$c) 0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m} \left[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} + g(z_1 - z_2) \right]$$

$$-\dot{Q}_{cv} = \dot{m} \left[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} \right]$$

$$\sum \dot{m}_i = \sum \dot{m}_e = \dot{m}$$

$$-\dot{Q}_{cv} = \dot{m}_i \left(h_i + \frac{V_i^2}{2} \right) + \sum \dot{m}_e \left(h_e + \frac{V_e^2}{2} \right)$$

$$e) \text{volumetric flow} = A V = \dot{m} V$$

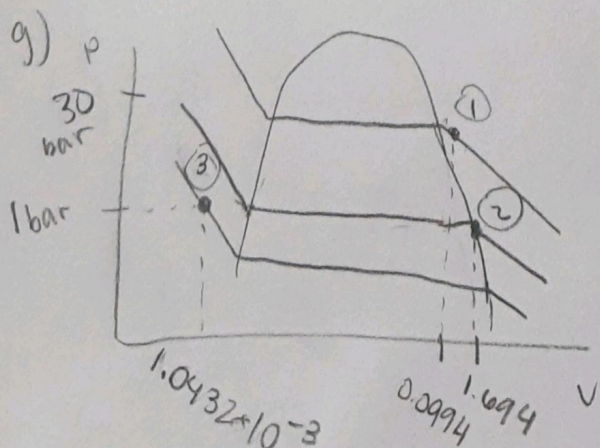
$$\dot{m} = \frac{A V}{V}$$

$$= 2 \frac{\text{kg}}{\text{s}} \left(1.694 \frac{\text{m}^3}{\text{kg}} \right)$$

$$= 3.388 \frac{\text{m}^3}{\text{s}}$$

$$\dot{Q}_{cv} = 2 \left(3230.9 + \frac{40^2}{2} \right) + 1 \left(2675.5 + \frac{100^2}{2} \right) + 1 \left(417.46 + \frac{90^2}{2} \right)$$

$$\dot{Q}_{cv} = 20186.76 \frac{\text{kJ}}{\text{s}} = 20186.76 \text{ kW}$$



3.1 a 5 / 5

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3.2 b 5 / 5

✓ - 0 pts Correct

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3.4 d 8 / 10

- **0 pts** Correct
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- **2 pts** KE term is not divided by 1000 (unit conversion)
- **10 pts** incorrect or not done
- **2 pts** not multiplied with mass outlets
- **1 pts** work sign is wrong
- ✓ - **2 pts** work is not added
- **2 pts** work unit is not converted
- **2 pts** mass is wrong
- **2 pts** h_1 is wrong
- **2 pts** h_3 is wrong
- **0 pts** [Click here to replace this description.](#)

3.5 e 3 / 5

- 0 pts Correct
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